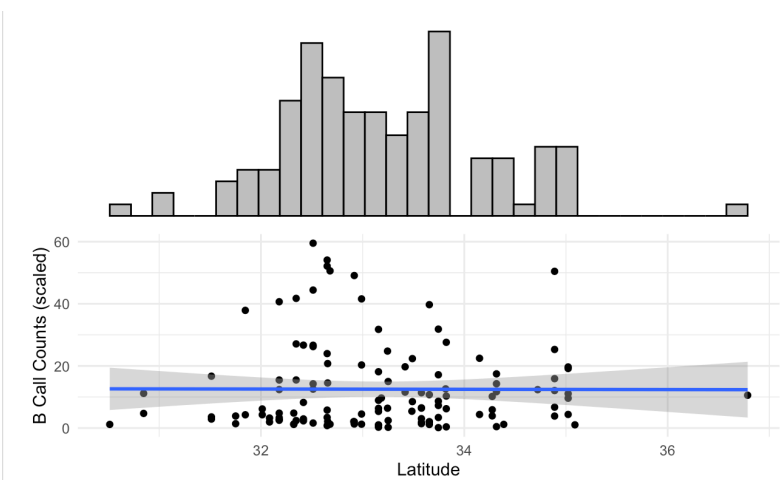
B Call Final Report

By: Samantha Su

Introduction. Tracking blue-whale "B" calls is valuable because these low-frequency vocalizations are strongly linked to breeding activity, allowing researchers to infer how the whales' habitat preferences shift through the year. We linked 2009-2023 CalCOFI whale-call records with CASE-STSE ocean data (temperature, salinity, current speed/direction at three depths), added latitude-longitude and season tags, then modeled blue-whale "B" call presence with a delta-gamma mixed model in **sdmTMB**. The model combines a logit term for presence/absence and a log term for call strength, uses a ~10 km spatial mesh, and treats season as a random effect. Diagnostics showed the model performed well, and the resulting heat maps highlight current breeding-activity "hotspots" to guide future monitoring.

Data Cleaning. We dropped the first two columns, kept only rows with a valid d_scaled value, and added blue_resp as the sum of a_scaled, b_scaled, and d_scaled.

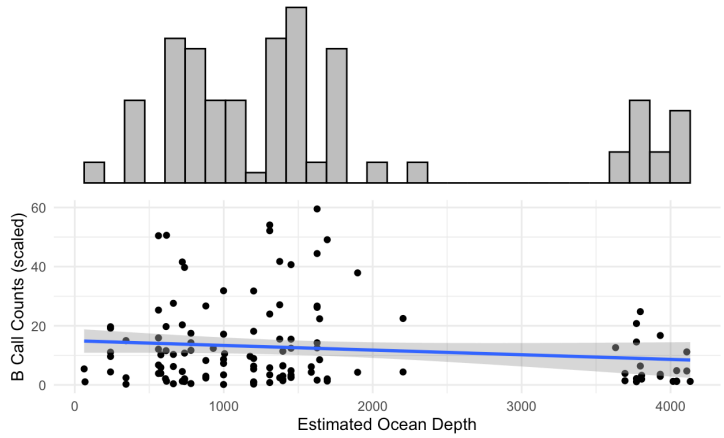
Exploratory Data Analysis. (Please note that the rest of the graphs are found in the b_call rmd file under modeling–here are just a couple that I wanted to mention). We conducted exploratory data analysis to investigate the relationship between the predictors environmental variables and response variable b_scaled in our spatiotemporal model. We observed that the response variable is highly zero-inflated, which may be attributed to several factors: (1) the absence of whales in the area, (2) the presence of whales that did not produce detectable sounds, or (3) the presence of loud background noise that masked the whale vocalizations. In order to explore the association between environmental variables and whale calls, we filtered zero value for visualizations.



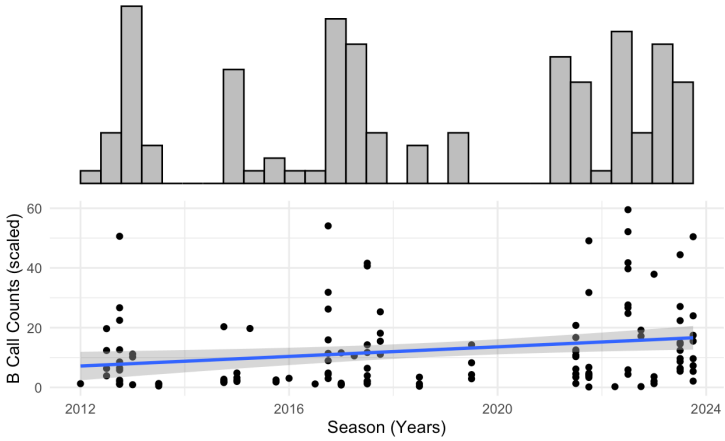
Southern latitudes (around 32°–34°) could be hotspots for B calls—perhaps due to habitat, prey concentration, or whale migration patterns.

The flat or slightly decreasing trend implies no strong latitudinal gradient in B call presence overall.

However, localized variability is high, as seen in the spread of values even at the same latitude.



We see two main depth zones where we recorded blue-whale B calls. In the mid-depth range (about 500–2,000 m), we picked up most of the loud, frequent calls, so whales seem to like spending time or breeding there. In very deep water (around 3,800–4,200 m), we took lots of samples but heard only a few calls, which means whales—or at least their vocalizations—are much rarer in that depth.



The bar chart at the top shows how many surveys happened each year. Effort was light from 2012–2015, grew a bit in 2016–2017, and jumped sharply after 2021. The scatterplot below maps every B-call count over time with a trend line. Most yearly counts are low, but some years, especially after 2021, have a few big spikes. That upward trend hints that B calls might be increasing overall, yet part of the rise could just be because we started sampling more often in recent years. In short, whale calls seem to have gone up since 2012, but uneven sampling makes it hard to know how much is real change versus better coverage.

Best Model. We used delta-gamma hurdle model, as it separately models the probability of presence and the magnitude of response when present. I began by creating a time-aware train/test split by putting everything except the most recent survey year into the training set and holding that final year out as a realistic “future” test set. I also set a seed to ensure any later random steps are reproducible. In this model, I set spatial = "on" and spatiotemporal = "iid" to capture both stable spatial patterns and independent seasonal variation in blue whale B-call activity. The spatial field accounts for consistent location-based differences—such as offshore versus coastal effects—while the "iid" setting allows each season to vary independently without assuming any temporal correlation. I also included

extra_time = unique(test_data$season) to ensure the model could make predictions for seasons that appear only in the test set. After fitting the model, I predicted on both the training and test data and evaluated performance using RMSE, range, and standard deviation. The model achieved a low test RMSE relative to the observed variability, indicating strong generalization and minimal overfitting. Out of all the models I tested, this one performed best, effectively capturing the spatiotemporal dynamics of B-call activity with an optimal balance of accuracy and simplicity.

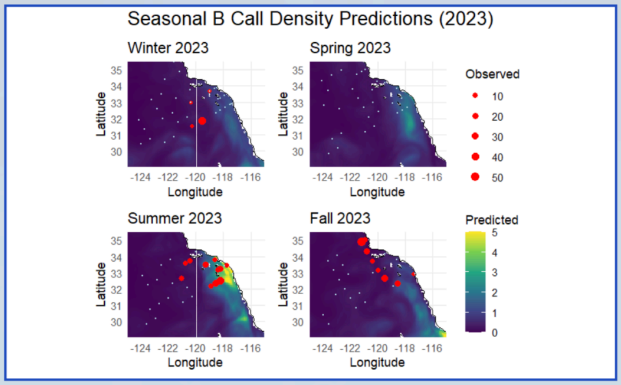


Fig. 7: Seasonal B Call Density Predictions

The spatiotemporal model for predicting blue whale **B call** presence with the lowest test RMSE is :

b_scaled ~ lat + lon + temperature_depth_55 + salinity_depth_55 + magnitude_depth_55 + theta_depth_55

- **Test RMSE: 5.9131**
- **Test Standard Deviation: 6.6932**

The heatmaps on the left show that B calls are predicted to occur **most frequently in the summer** (particularly near the coast), indicating strong model prediction this season, and **least in the spring**, showing that there are low predicted activities. Fall predictions are more moderate, although there are more observed calls than predicted, so there might be a spatial mismatch.

Conclusion. This model was the best at predicting where blue whale B calls occurred, with a low error rate that suggests it captured the main patterns in the data well. It showed that B calls were most frequent in the summer, especially near the southern California coast, which may indicate important habitat during that season. In contrast, very few calls were predicted in the spring. Fall predictions were more mixed—while the model predicted moderate activity, the actual number of calls was higher in some areas, suggesting it may have missed some important spots.

In the future, we could improve the model by testing whether call activity is related across seasons (not just independent), using a finer spatial grid, or including environmental data from deeper parts of the ocean.